$r(t) = 2\sin\left(\frac{t}{2}\right)i + 2\cos\left(\frac{t}{2}\right)j$

Find: $v(t) = \cos\left(\frac{t}{2}\right)i - \sin\left(\frac{t}{2}\right)j$

$a(t) = -\frac{1}{2}\sin\left(\frac{t}{2}\right)i - \frac{1}{2}\cos\left(\frac{t}{2}\right)j$

speed: $\sqrt{\left(\cos\left(\frac{t}{2}\right)\right)^2 + \left(-\sin\left(\frac{t}{2}\right)\right)^2}$

$= 1$

Velocity is r′(t), acceleration is r″(t), speed is ‖r′(t)‖. Here v = cos(t/2)i − sin(t/2)j, so the speed is √(cos² + sin²) = 1.

Let C be a smooth curve represented by r on interval I

(r(t) is a space curve)

Setup: C is a smooth curve traced by r(t) on an interval I. Smooth means r′(t) is continuous and never the zero vector.

Unit Tangent Vector

$$T(t) = \frac{r'(t)}{\|r'(t)\|}$$

The unit tangent vector T(t) = r′(t)/‖r′(t)‖ points in the direction of motion and always has length 1.

$r(t) = t\,i + t^2 j$

$T(t) = \frac{r'(t)}{\|r'(t)\|} = \frac{1}{\|r'(t)\|} \cdot r'(t)$

$r'(t) = 1\,i + 2t\,j$

$\|r'(t)\| = \sqrt{(1)^2 + (2t)^2} = \sqrt{1 + 4t^2}$

$T(t) = \frac{1i + 2tj}{\sqrt{1+4t^2}} = \frac{1}{\sqrt{1+4t^2}}(1i + 2tj)$

$T(1) = \frac{1}{\sqrt{5}}(1i + 2j)$

$= \frac{1}{\sqrt{5}}i + \frac{2}{\sqrt{5}}j$

For r(t) = t i + t² j: r′ = i + 2t j and ‖r′‖ = √(1 + 4t²), so T(t) = (i + 2t j)/√(1 + 4t²). At t = 1, T = (1/√5)i + (2/√5)j.

$r(t) = 2\cos(t)\,i + 2\sin(t)\,j + k$

Find $T\left(\frac{\pi}{4}\right)$

$r'(t) = -2\sin(t)i + 2\cos(t)j + 0$

$\|r'(t)\| = 2$

$T(t) = -\sin(t)i + \cos(t)j$

$T\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}j$

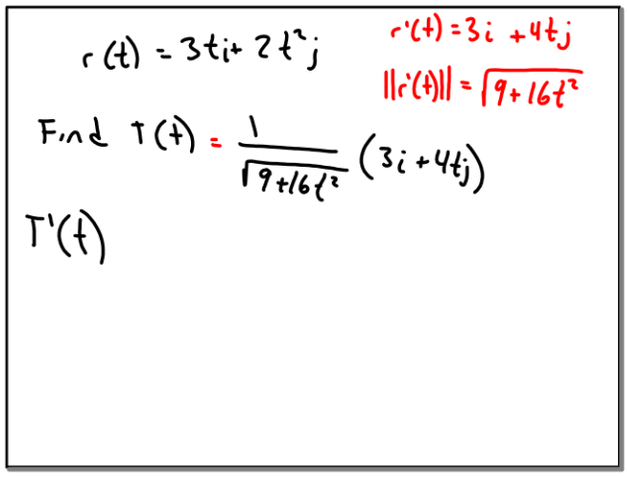
r′(t) = −2sin t i + 2cos t j has constant length 2, so T(t) = −sin t i + cos t j and T(π/4) = −(√2/2)i + (√2/2)j.

Principal Unit Normal Vector

$$N(t) = \frac{T'(t)}{\|T'(t)\|} \neq \frac{r''(t)}{\|r''(t)\|}$$

The principal unit normal N(t) = T′(t)/‖T′(t)‖ points the way the curve bends. Normalize T′, not r″: r″/‖r″‖ is generally a different vector.

Produced in 2026



Start of a normal-vector example: r′ = 3i + 4t j and ‖r′‖ = √(9 + 16t²), so T(t) = (3i + 4t j)/√(9 + 16t²). T′(t) is finished in the 10/7 notes.